



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2013

BIOLOGY

Higher 1

Paper 2

8875/02

Monday, 16 September

2 hours

Candidate Name: _____

Civics Group: _____

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions

Section B

Answer any **one** question

At the end of the examination, fasten the answer papers securely and hand up separately.
The number of marks is given in brackets [] at the end of each question or part question.

For examiner's use	
Section A	
1	
2	
3	
4	
Section B	
5 / 6	
TOTAL	

This document consists of **14** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

- 1** Oxygen is an essential molecule for eukaryotic organisms as it is needed for the generation of ATP for cellular processes. In mammals, oxygen is carried by haemoglobin. Haemoglobin consists of 4 globin subunits, and each subunit has a prosthetic group called the heme group, where an iron ion binds to an oxygen molecule.

- (a)** Name the precise type of bond between the monomers in haemoglobin and describe how it is formed. Use an annotated diagram in the space below to illustrate your answer.

Name of bond:

Formation:

.....

.....

.....

Diagram:

[4]

[Turn over

Fig. 1.1 shows the structure of one of the haemoglobin subunits with the prosthetic group.

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(b) Describe the level of protein structure seen in each globin subunit.

.....

.....

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.....

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..... [3]

[Turn over

Haemoglobin has the ability to bind to other molecules through the same heme group. Fig. 1.2 shows the affinities of different types of haemoglobin for molecule X at different concentrations of the molecule (reflected by its partial pressure).

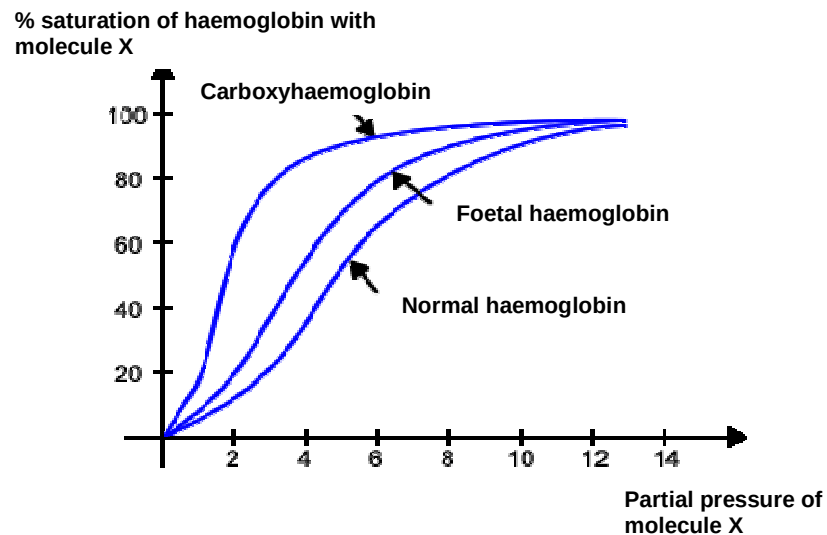


Fig. 1.2

Another example of a molecule that binds to haemoglobin is carbon monoxide (CO). It is commonly found in cigarette smoke, smoke from car exhaust pipes etc., and when bound to haemoglobin, it forms a molecule called carboxyhaemoglobin. Small amounts of CO can dramatically reduce haemoglobin's ability to transport oxygen. When inspired air contains as low as 0.02% CO, headache and nausea can set in. 0.1% CO, the person will become unconscious.

- (c) (i) Suggest whether haemoglobin has a higher or lower affinity for CO as compared to oxygen.

..... [1]

- (ii) Explain the physiological effects of CO poisoning in a human.

..... [2]

(iii) Suggest how CO poisoning can be treated.

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.....

..... [1]

[Total: 11]

[Turn over

- 2 Antigens are found on the human red blood cell and are used to determine the human ABO blood group system. Type and presence of antigens are determined by three alleles, I^A , I^B and I^O . Alleles I^A and I^B are co-dominant and both alleles are dominant to allele I^O .

(a) Define the term co-dominance using the example of the ABO blood group.

.....

.....

.....

..... [2]

ABO haemolytic disease of the newborn refers to blood disorders that arise when the blood groups of the mother and newborn child do not match. It usually happens during a mother's second or subsequent pregnancy.

Antibodies from the mother can pass through the placenta and bind to the child's red blood cells. This leads to cell lysis, resulting in anaemia. Table 2.1 shows the three different situations where blood group incompatibility can arise between a mother and her child

Table 2.1

	Blood group		
	Situation 1	Situation 2	Situation 3
Mother	O	A	B
Child	A or B	B	A

A man with blood type A married a woman with blood type O.

(b) Use genetic diagrams to show the genotypes of the children that the couple can have.

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Use

[4]

[Turn over

- (c) Suggest a possible treatment if the newborn suffers from severe anaemia due to ABO haemolytic disease.

.....
 [1]

The ABO blood group system behaves according to Mendelian laws and the environment has no effect on the kind of blood group a person has.

In some other cases, the phenotype of an organism is dependent on its environment. An example is a species of hydrangea, *Hydrangea macrophylla*, where the colour of the flowers changes according to the conditions of the soil. Table 2.2 shows the effects of varying pH and aluminium concentration on the colour of hydrangea flowers.

Table 2.2

Factors affecting colour of hydrangea		Aluminium Concentration	
		Low	High
Soil pH	5.5	Blue	Blue
	6.0	Pink	Blue

With reference to Table 2.2,

- (d) (i) describe the effect of soil pH and aluminium concentration on flower colour.

.....

 [2]

- (ii) suggest how the colour of hydrogea can be useful.

.....
 [1]

[Total: 10]

- 3** Fructose 1, 6-bisphosphatase is a crucial enzyme involved in the process of regenerating ribulose biphosphate. Studies using etiolated spinach seedlings show residual fructose 1, 6-bisphosphatase activity, which sharply rises under illumination.

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Fig. 3.1 shows a section of the organelle where fructose 1, 6-bisphosphatase can be obtained.

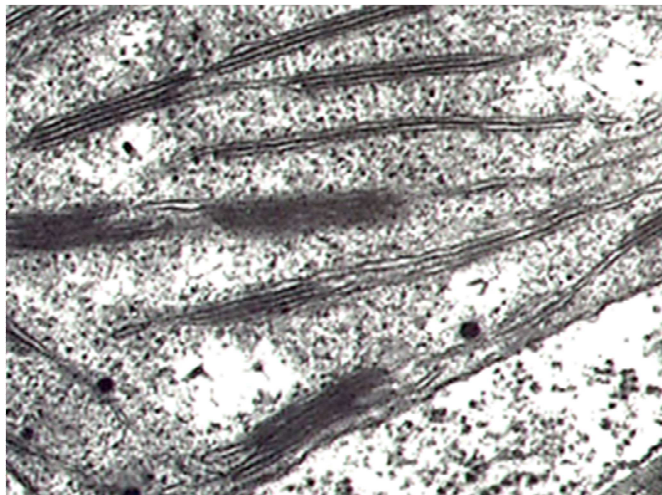


Fig. 3.1

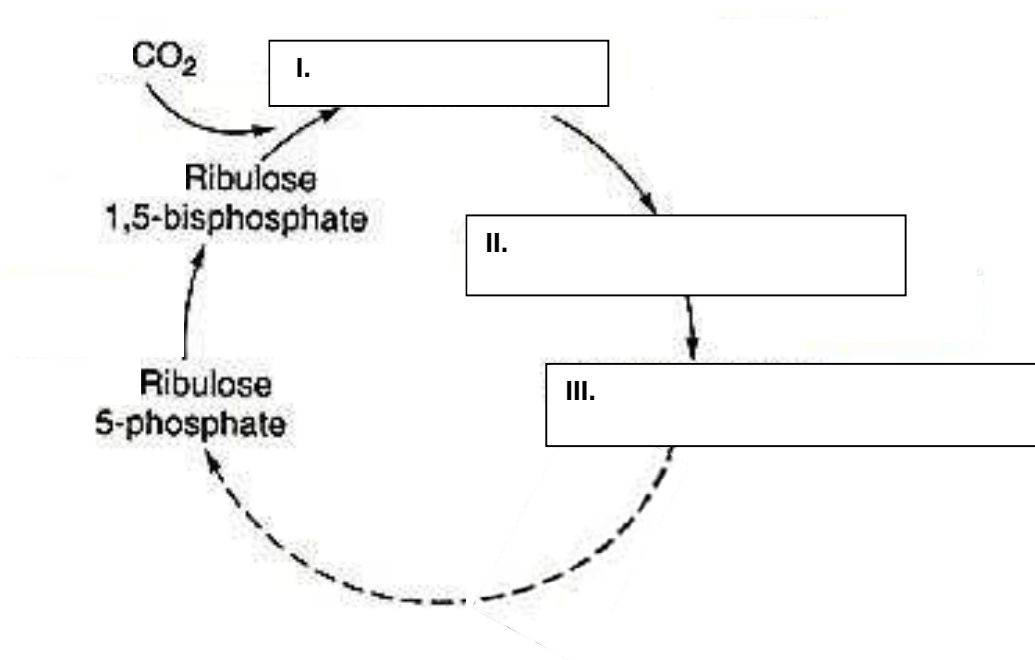
- (a) (i)** State the precise location of fructose 1, 6-bisphosphatase activity.

.....

- (ii)** Annotate in Fig. 3.1 your answer for **(a) (i)**.

[1]

(b) The following cycle also occurs in the same organelle.



(i) Fill in the blanks to complete the cycle.

[3]

(ii) Draw the structure of the storage form of the sugar produced from the cycle in the space below. Indicate the different types of bonds present.

[2]

[Turn over

(iii) Explain **two** key functions of the cycle, other than the synthesis of sugar.

.....

.....

.....

..... [2]

Cycloheximide is an inhibitor of protein biosynthesis in eukaryotic organisms. The rise of fructose 1, 6-bisphosphatase activity under illumination has been found to be strongly inhibited by cycloheximide, but not by lincocin. Lincocin is a 70S ribosome inhibitor.

(c) Explain if the gene coding for fructose 1, 6-bisphosphatase is found in the nucleus or the chloroplast.

.....

.....

.....

..... [2]

[Total: 10]

- 4 In the early studies of relationships between vertebrates, the embryos of various species were observed. The key stages of development (Stage I to III) in some of these species are illustrated in Fig. 4.1.

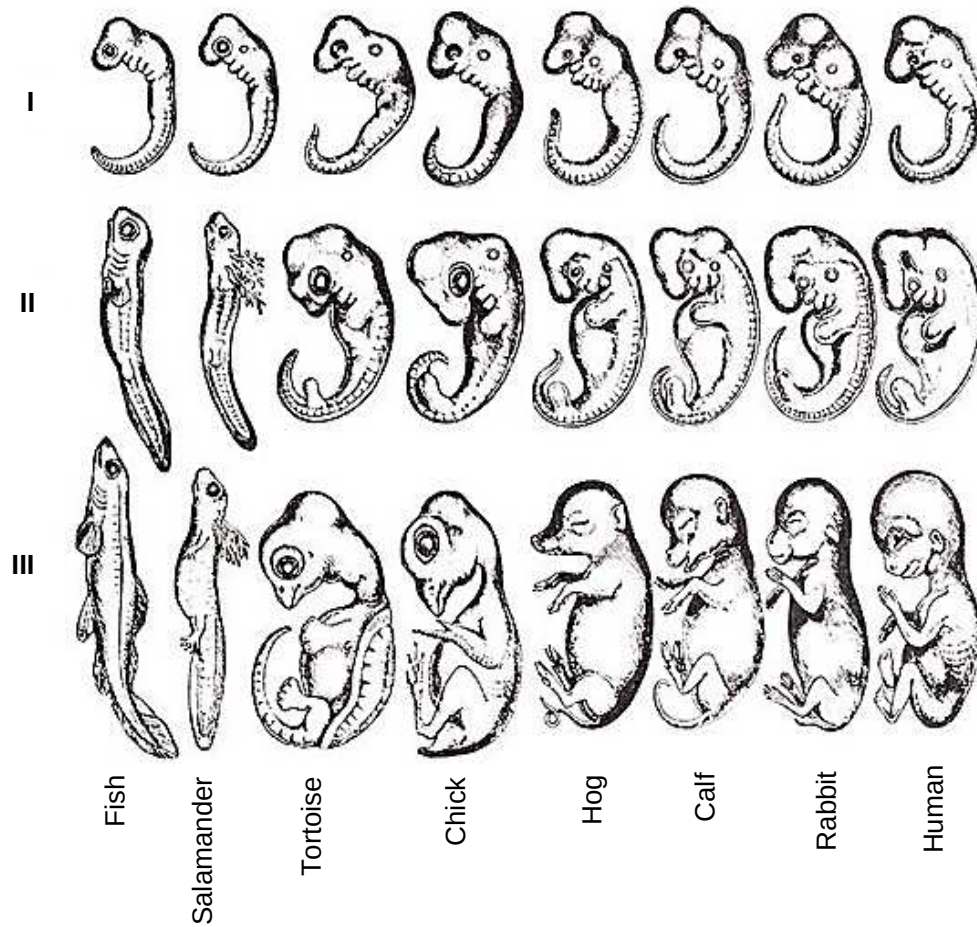


Fig. 4.1

- (a) With reference to Fig. 4.1,
- (i) state **two** features that are common to all the embryos.

[2]

[4]

..... [2]

..... [1]

[Turn over

SECTION B

Free response question

Answer one question.

Write your answers to this question on the separate answer paper provided.

Your answers

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Relate the structure of the nucleus to its roles in protein synthesis. [7]

(b) Inhibitors such as acrylamide have been known to affect enzymes involved in respiration. Describe how you would carry out an investigation to determine if acrylamide is a competitive or non-competitive inhibitor of phosphofructokinase in a named organism. [8]

(c) Describe how meiosis and fertilization result in genetic variation. [5]

[Total: 20]

OR

6 **(a)** Outline the procedure for the production of human insulin protein using bacterial transformation and only **one** agar plate containing ampicillin and X-gal. [8]

(b) Discuss the benefits and possible issues of rational drug design using the information obtained from human genome project. [6]

(c) Explain the normal functions of embryonic and blood stem cells in a living organism and distinguish the unique features between them. [6]

[Total: 20]